

Sagar Singh Bhola

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EDUCATION

Georgia Institute of Technology

Master of Science in Computer Science, Specialization in Artificial Intelligence

Expected December 2028

University of Michigan—Ann Arbor

Ann Arbor, MI

Bachelor of Science in Computer Science | GPA: 3.75

Aug. 2021 – Dec. 2025

- **Coursework:** Data Structures and Algorithms, Computer Networks, Computer Security, Computer Architecture

TECHNICAL SKILLS

Languages/Technologies: Python, C/C++, JavaScript, TypeScript, SQL, Java, HTML/CSS

Frameworks/Libraries: React, Next.js, Vue, Node.js, Express.js, Flask, FastAPI, Pandas, PyTorch, OpenCV

Tools/Platforms: MySQL, Docker, AWS, Git, Postman, Playwright, Figma

EXPERIENCE

Data Engineering Co-op

May 2025 – Present

Ciena Healthcare

Southfield, MI

- Automated Michigan Medicaid cost-report preparation and validation across 51 nursing facilities, supporting an 84% first-pass acceptance rate and helping avoid approximately \$100K in external preparation costs.
- Built a Python ETL pipeline with Pandas, pdfplumber, regex, and OpenPyXL to process 2,000+ Medicaid reimbursement PDFs across four fiscal years, reducing a multi-day historical backfill to under two hours.
- Developed Python and SQL reconciliation pipelines comparing PointClickCare census and patient-in-bed data with general-ledger records, automatically flagging cross-system discrepancies before monthly financial close.

PROJECTS

Networthy | *Next.js, TypeScript, Node.js, Express.js, PostgreSQL, Playwright*

Aug. 2026 – Present

- Built a creator-finance platform that consolidates earnings data from multiple monetization sources into a unified revenue model and dashboard.
- Architected proprietary API connectors and Playwright automation to retrieve, normalize, and ingest platform revenue data using Node.js and AWS.
- Built persistent creator profiles to ground AI-generated content ideas in creator-specific context, enabling personalized ideation and content planning.

Vision-Based Autonomous Driving Agent | *Python, PyTorch, OpenCV, CUDA, C++*

Aug. 2026 – Present

- Built a closed-loop behavior-cloning agent that predicts steering, throttle, and braking from first-person visual input, enabling autonomous highway driving in Assetto Corsa traffic.
- Collected and synchronized 80,000+ gameplay frames with Thrustmaster T300RS control inputs across 15+ driving sessions, using session-level validation to reduce temporal leakage.
- Developed a real-time 30 FPS perception-to-control pipeline with GPU-accelerated inference, connecting model outputs to virtual pedals and physical steering through a C++ force-feedback controller.

AI Invoice Processor | *Python, Next.js, FastAPI, Ollama, Qwen, Docling*

Apr. 2026 – May 2026

- Built a privacy-focused invoice processor that converts uploaded documents into structured accounting data using a locally hosted language model and schema-constrained extraction.
- Developed an end-to-end workflow for invoice ingestion, extraction, validation, human review, and persistence, reducing average entry time from 4 minutes to 30 seconds.
- Implemented stateful accounts-payable review with embedded PDF previews, editable invoice fields and line items, and persistent drafts for human-in-the-loop correction.

YesNoFingers | *Python, OpenAI Whisper, AWS Translate, AWS EC2*

May 2025 – Present

- Built a cloud-based medical speech-translation system with the University of Michigan Language Resource Center, supporting real-time translation across 40+ languages using Whisper and AWS Translate.
- Engineered a Python pipeline on AWS EC2 for audio validation, transcription, terminology-aware preprocessing, translation, and API responses, achieving approximately three-second end-to-end processing per recording.
- Piloted the system with five clinicians at a Detroit-area free clinic, using their feedback to improve API reliability, input validation, and failure handling.